

Survey-weighted estimation with MORIE

Overview

Many MORIE analyses use Statistics Canada Public Use Microdata Files (CCS, CSADS, CSUS, CADS, CPADS), which require complex-sample design weights for valid inference. This vignette covers the sampling and weighting helpers that wrap the same machinery.

Stratified, cluster, and PPS sampling

```
library(morie)
set.seed(11)

n <- 1000
df <- data.frame(
  id      = seq_len(n),
  region = sample(c("ON", "QC", "BC", "AB"), n, replace = TRUE),
  age     = round(rnorm(n, 40, 12)),
  weight  = runif(n, 0.5, 2.0)
)

# Stratified random sample within region.
strat <- stratified_sample(df, strata_col = "region", n_per_stratum = 50)

# Single-stage cluster sample on region (all units in sampled
# clusters are kept).
clust <- cluster_sample(df, cluster_col = "region", n_clusters = 2)

dim(strat)
#> [1] 200  5
dim(clust)
#> [1] 505  5
```

Calibration weights and design weights

`compute_design_weights()` and `calibration_weights()` produce weights consistent with a known sampling scheme and known population marginals.

```
# Design weights from a stratified scheme.
# Treat `region` as the stratum; supply known population sizes per stratum.
pop_sizes <- c(ON = 14000000, QC = 8500000, BC = 5200000, AB = 4400000)
design <- compute_design_weights(
```

```

df,
  strata_col      = "region",
  population_sizes = pop_sizes
)
head(design)
#> [1] 34000.00 34000.00 18107.00 55555.56 18107.00 55555.56

```

Effective sample size and design effect

```

# Effective sample size given a weight vector.
ess <- effective_sample_size(df$weight)
ess
#> [1] 897.9353

# Design effect: how much variance inflates from the weighting.
deff <- design_effect(df$weight)
deff
#> [1] 1.113666

```

The effective sample size answers the practical question “how many independent observations does this weighted sample contain?” The design effect translates between unweighted-equivalent and complex-sample variances.

Bootstrap variance with weights

```

# Bootstrap a weighted statistic --- here, the weighted mean of `age`.
boot <- bootstrap_sample(
  df,
  statistic = function(x) stats::weighted.mean(x$age, x$weight),
  n_bootstrap = 100
)
str(boot, max.level = 1)
#> List of 5
#> $ estimate      : num 39.9
#> $ se            : num 0.425
#> $ ci_lower      : Named num 39.1
#> ..- attr(*, "names")= chr "2.5%"
#> $ ci_upper      : Named num 40.6
#> ..- attr(*, "names")= chr "97.5%"
#> $ distribution: num [1:100] 40.3 40 39.1 39.9 39.6 ...

```

Where to go next

- For complex-survey causal inference (survey-weighted ATE/AIPW), see the **causal-inference** vignette — the same **estimate_*** functions accept a **weights** argument.
- For Statistics Canada PUMF acknowledgments and citation requirements, see the package README.